SMIT — Batch 22

JavaScript Assignment 02: Conditions & Logical Operators

Course:	Web & Mobile App Development
Subject:	JavaScript
Assignment Number:	02
Reference Book:	A Smarter Way to Learn JavaScript
Chapters:	10–14 (If Statements, Comparison Operators, Else If, Logical Operators, Nested If)
Total Marks:	100
Duration:	5 Days
Submission:	GitHub Repository

Objective

This assignment is designed to improve your logical thinking and decision-making skills using JavaScript conditional statements and logical operators.

By completing this assignment, you will learn how to solve real-world problems using:

- `if`, `else if`, `else`
- Comparison operators (>, <, >=, <=, ==, ===, `!=`, `!==`)
- Logical operators (`&&`, `||`, `!`)
- Nested conditional statements
- User input using `prompt()`
- Mathematical expressions and calculations

General Instructions

1. Solve all questions using JavaScript.
2. Use `prompt()` to collect user input wherever required.
3. Display results using `alert()`, `console.log()`, or HTML output.
4. Use meaningful variable names and proper indentation.
5. Use `if`, `else if`, `else`, and logical operators wherever appropriate.
6. Do not use loops, arrays, functions, or objects.
7. Do not copy code from classmates or AI tools.
8. Push your complete assignment to GitHub.
9. Be prepared to explain your logic during class.

Section A — Basic Conditions (20 Marks)

Question 1: Positive, Negative or Zero (5 Marks)

Write a JavaScript program that takes a number from the user and determines whether it is:

- Positive
- Negative
- Zero

Example Input:

```
Enter a number: -8
```

Expected Output:

```
-8 is a negative number.
```

Question 2: Even or Odd Number (5 Marks)

Ask the user to enter an integer and determine whether it is even or odd using the modulus operator.

Example Input:

```
Enter a number: 17
```

Expected Output:

```
17 is an odd number.
```

Question 3: Voting Eligibility (5 Marks)

Create a program that asks the user for their age.

Conditions:

- Age 18 or above: Eligible to vote
- Below 18: Not eligible to vote
- Negative age: Invalid age

Example Input:

```
Enter your age: 16
```

Expected Output:

```
You are not eligible to vote.
```

Question 4: Find the Largest Number (5 Marks)

Take three numbers from the user and find the largest number using conditional statements.

Example Input:

```
First Number: 25
Second Number: 80
Third Number: 45
```

Expected Output:

```
80 is the largest number.
```

If multiple numbers share the maximum value, display that maximum.

Section B — Comparison & Logical Operators (25 Marks)

Question 5: Login Authentication (5 Marks)

Create a simple login system using fixed credentials.

Credentials:

```
Username: admin
Password: admin123
```

Ask the user to enter their username and password.

Conditions:

- Both correct: Login Successful
- Username incorrect: Invalid Username
- Username correct but password incorrect: Invalid Password

Example Output:

```
Login Successful! Welcome Admin.
```

Question 6: Student Grade Calculator (5 Marks)

Ask the user to enter their percentage.

Determine the grade using the following criteria:

Percentage	Grade
90–100	A+
80–89	A
70–79	B
60–69	C
50–59	D
Below 50	Fail

Conditions:

- Percentage must be between 0 and 100.
- Display an error message for invalid percentages.

Example Input:

```
Enter percentage: 85
```

Expected Output:

```
Your Grade is A.
```

Question 7: Discount Eligibility (5 Marks)

A shopping store offers discounts based on the total purchase amount.

Purchase Amount	Discount
Rs. 10,000 or above	20%
Rs. 5,000–9,999.99	10%
Rs. 2,000–4,999.99	5%
Below Rs. 2,000	No discount

Ask the user to enter their total purchase amount.

Calculate and display:

- Original amount
- Discount percentage
- Discount amount
- Final payable amount

Example Input:

```
Purchase Amount: 6000
```

Expected Output:

```
Original Amount: Rs. 6000
Discount: 10%
Discount Amount: Rs. 600
Final Amount: Rs. 5400
```

Reject negative purchase amounts.

Question 8: Leap Year Checker (5 Marks)

Ask the user to enter a year.

Determine whether it is a leap year.

Rules:

- A year divisible by 400 is a leap year.
- A year divisible by 100 but not 400 is not a leap year.
- Otherwise, a year divisible by 4 is a leap year.

Example Input:

```
Enter Year: 2024
```

Expected Output:

```
2024 is a Leap Year.
```

Question 9: Job Eligibility Checker (5 Marks)

A company is hiring junior developers.

Requirements:

- Age must be between 18 and 30 (inclusive).
- Minimum experience: 1 year.
- JavaScript knowledge is required.

Ask the user for:

- Age
- Years of experience
- JavaScript knowledge (`yes` or `no`)

Expected Output:

```
Congratulations! You are eligible to apply.
```

Otherwise:

```
Sorry! You do not meet the requirements.
```

Reject negative ages or experience values.

Section C — Nested Conditions & Real-World Problems (35 Marks)

Question 10: ATM Withdrawal System (10 Marks)

Create an ATM withdrawal program.

Initial Balance: Rs. 25,000

Ask the user for:

- PIN
- Withdrawal amount

Correct PIN: `1234`

Conditions:

1. Validate the PIN.
2. Withdrawal amount must be greater than zero.
3. Withdrawal amount must not exceed the available balance.
4. Withdrawal amount must be a multiple of Rs. 500.
5. If all conditions are satisfied, deduct the amount from the balance.

Example Input:

```
Enter PIN: 1234
Enter Amount: 5000
```

Expected Output:

```
Withdrawal Successful!
Withdrawn Amount: Rs. 5000
Remaining Balance: Rs. 20000
```

Display appropriate error messages for invalid transactions.

Question 11: Electricity Bill Calculator (10 Marks)

An electricity company charges customers based on monthly consumption.

Units Consumed	Rate per Unit
0–100	Rs. 10
101–200	Rs. 15
201–300	Rs. 20
Above 300	Rs. 25

Billing Rule: Apply one rate to all units based on the total consumption bracket (not progressive slab billing).

Ask the user to enter consumed units.

Calculate:

- Electricity charges
- Fixed service charges: Rs. 150
- Tax: 5% of electricity charges
- Total payable bill

Example Input:

```
Units Consumed: 250
```

Expected Output:

```
Electricity Charges: Rs. 5000
Service Charges: Rs. 150
Tax: Rs. 250
Total Bill: Rs. 5400
```

Reject negative or non-numeric units.

Question 12: Student Result System (10 Marks)

Ask the user to enter marks for five subjects:

- English
- Mathematics
- Science
- Computer
- Urdu

Each subject has a maximum of 100 marks.

Requirements:

1. Calculate total obtained marks.
2. Calculate percentage.
3. Determine grade using the grading criteria from Question 6.
4. A student must score at least 40 marks in every subject to pass.
5. If any subject has marks below 40, the overall result must be Fail, regardless of percentage.
6. Reject marks below 0 or above 100.

Expected Output:

```
Total Marks: 500
Obtained Marks: 425
Percentage: 85%
Grade: A
Result: PASS
```

Question 13: Traffic Signal System (5 Marks)

Ask the user to enter a traffic signal color.

Conditions:

- Red -> Stop
- Yellow -> Get Ready
- Green -> Go
- Any other input -> Invalid Signal

The program should accept both uppercase and lowercase inputs.

Example:

```
Enter Signal: RED
Output: Stop
```

Section D — Problem-Solving Challenge (20 Marks)

Question 14: Cinema Ticket Booking System (10 Marks)

A cinema offers different ticket prices based on age.

Age Group	Ticket Price
Below 5 years	Free
5–12 years	Rs. 300
13–59 years	Rs. 700
60 years and above	Rs. 400

Additional Rules:

- Weekend surcharge: Rs. 100 per paid ticket.
- Children below 5 do not pay the surcharge.
- Negative age is invalid.

Ask the user for:

- Age
- Day type (`weekday` or `weekend`)

Calculate and display the final ticket price.

Example Input:

```
Age: 25
Day Type: weekend
```

Expected Output:

```
Base Ticket Price: Rs. 700
Weekend Surcharge: Rs. 100
Final Ticket Price: Rs. 800
```

Question 15: Employee Salary & Bonus System (10 Marks)

A company calculates employee bonuses based on years of service.

Years of Service	Bonus
Less than 1 year	No bonus
1–3 years	5%
More than 3–5 years	10%
More than 5 years	15%

Additional Rules:

- Employees with a salary of Rs. 100,000 or above pay 5% tax on their basic salary.
- Other employees pay 2% tax on their basic salary.
- Final salary = Basic Salary + Bonus - Tax.
- Salary and years of service cannot be negative.

Ask the user for:

- Basic salary
- Years of service

Example Input:

```
Basic Salary: 80000
Years of Service: 4
```

Expected Output:

```
Basic Salary: Rs. 80000
Bonus: Rs. 8000
Tax: Rs. 1600
Final Salary: Rs. 86400
```

Submission Guidelines

Repository Name: `SMIT-Batch22-JS-Assignments`

Folder Structure:

```
SMIT-Batch22-JS-Assignments/
|-- Assignment-01/
`-- Assignment-02/
    |-- index.html
    |-- question-01.js
    |-- question-02.js
    |-- ...
    |-- question-15.js
    `-- README.md
```

Submission Requirements:

- Complete all 15 questions.
- Push the code to GitHub.
- Add student name and batch number in the README.
- Include screenshots of at least five program outputs.
- Submit the GitHub repository link.

Marking Criteria

Criteria	Marks
Correct Output	35
Conditional Logic	30
Input Validation & Edge Cases	20
Code Quality & Readability	10
GitHub Submission	5
Total	**100**

Important Instructions

- Do not use arrays, loops, functions, objects, or advanced JavaScript methods.
- Use only concepts taught in class.
- Use appropriate comparison and logical operators.

- Validate user inputs before performing calculations.
- Write readable and properly indented code.
- Students may be asked to explain any question during class.

Good luck, Batch 22!

Remember: Programming is not about memorizing syntax. It is about understanding problems and building solutions.